

## Introduction to Gaussian elimination

Recall from the previous lecture that given a linear system with  $m$  equations in  $n$  variables we can write it as one matrix equation. When solving these systems of linear equations, we shall find it useful to combine the coefficient matrix of the linear system and vector of constants into one  $m \times (n + 1)$  matrix called the augmented matrix of the linear system.

That is, given a linear system below:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

Its associated augmented matrix is:

$$\left( \begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right)$$

**Example** Given a linear system below

$$\begin{aligned} 2x_1 - 3x_2 + 4x_3 &= 1 \\ 4x_2 - x_3 &= 3 \\ x_1 - x_2 + 2x_3 &= 2 \end{aligned}$$

its augmented matrix is

$$\left( \begin{array}{ccc|c} 2 & -3 & 4 & 1 \\ 0 & 4 & -1 & 3 \\ 1 & -1 & 2 & 2 \end{array} \right)$$

**Definition 17.** Two systems of linear equations in the same variables are called equivalent if the two systems have the same solution set.

We now seek operations on a linear system that can help ‘transform’ it into a simpler linear system to solve that has the same solution set as the linear system we began with.

The following operations are reversible and do not change the solution set.

- Interchanging two equations on our linear system.
- Multiplying an equation by a non-zero scalar.
- Replacing an equation by its sum with another equation.

The above are called the elementary operations on linear systems, and given the relationship between systems of linear equations and their associated augmented matrices, these elementary operations will correspond to row operations on our augmented matrices often called elementary row operations.

Thus in solving linear systems we shall make use of the following row operations on their associated augmented matrices.

- Interchange two rows.
- Multiply a row by a non-zero scalar.
- Replace a row by its sum with another row.

*Note that combining the last two we can replace a row by its non-zero scalar multiple added to another non-zero scalar multiple of a different row.*

Let us now try to solve our linear system in the above example, that is,

$$\begin{aligned}2x_1 - 3x_2 + 4x_3 &= 1 \\4x_2 - x_3 &= 3 \\x_1 - x_2 + 2x_3 &= 2\end{aligned}$$

We shall begin with its augmented matrix and perform elementary row operations.

$$\begin{pmatrix} 2 & -3 & 4 & 1 \\ 0 & 4 & -1 & 3 \\ 1 & -1 & 2 & 2 \end{pmatrix}$$

In the previous lecture, we saw that systems of linear equations in a triangular form are easy to solve using back-substitution. Let us then **attempt** to transform the above augmented matrix into triangular form. We shall attempt to do so in a finite number of steps using our elementary row operations.

Let us denote by  $R_1, R_2$  and  $R_3$  the first row, second row and third row, respectively.

1. If we replace  $R_3$  with  $2R_3 - R_1$ , we get

$$\begin{pmatrix} 2 & -3 & 4 & 1 \\ 0 & 4 & -1 & 3 \\ 0 & 1 & 0 & 3 \end{pmatrix}$$

2. Now, on the above, if we replace  $R_3$  with  $4R_3 - R_2$ , we get

$$\begin{pmatrix} 2 & -3 & 4 & 1 \\ 0 & 4 & -1 & 3 \\ 0 & 0 & 1 & 9 \end{pmatrix}$$

Notice that our **attempt** to get it into triangular form worked and the above is an augmented matrix corresponding to the system below.

$$\begin{aligned}2x_1 - 3x_2 + 4x_3 &= 1 \\4x_2 - x_3 &= 3 \\x_3 &= 9\end{aligned}$$

Since we have only used elementary row operations, the two systems must be equivalent and thus have the same solution. Solve the latter using back-substitution and hence obtaining the solution to our original system ( $x_1 = -13, x_2 = 3, x_3 = 9$ ).

Can we always transform any linear system into triangular form? No, but fortunately, we shall be able to transform it into another simpler form called the row echelon form, in particular triangular form is in row echelon form... We have discussed a few examples in class but let us continue the story in the next lectures...